Scientists' Exploration and Atypical Knowledge Combination

— Measuring research activities using deep learning

최세영

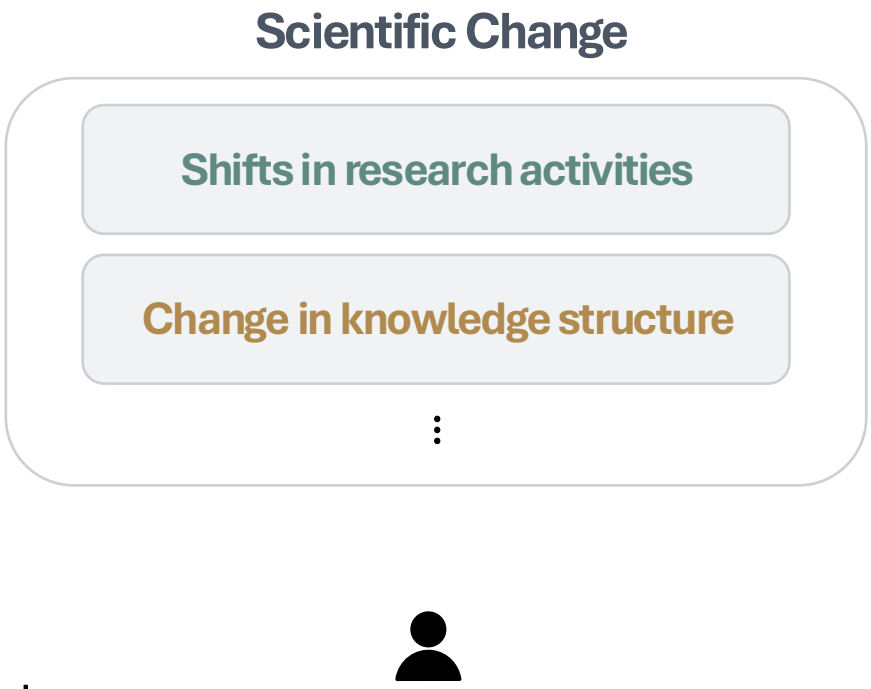
서울대학교 기술경영경제정책전공

Scientific Change

- **Scientific change**
  - **Changes over time in the organization of research activity and scientific knowledge.**
  - Scientific developments involve *social and cognitive changes* that should be distinguished analytically (Leydesdorff).

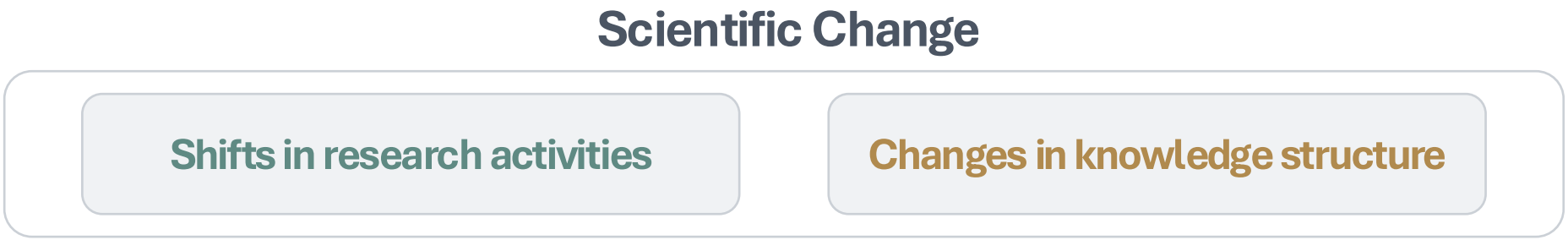
  - We focuses on **two interrelated aspects of scientific change**:
    - Shifts in the distribution of research activity
    - Changes in the structure of scientific knowledge

  - We examine ***scientists' behaviors*** that contribute to these changes.



Leydesdorff, *The Challenge of Scientometrics* (2001)

Scientific Change Appears in Two Aspects

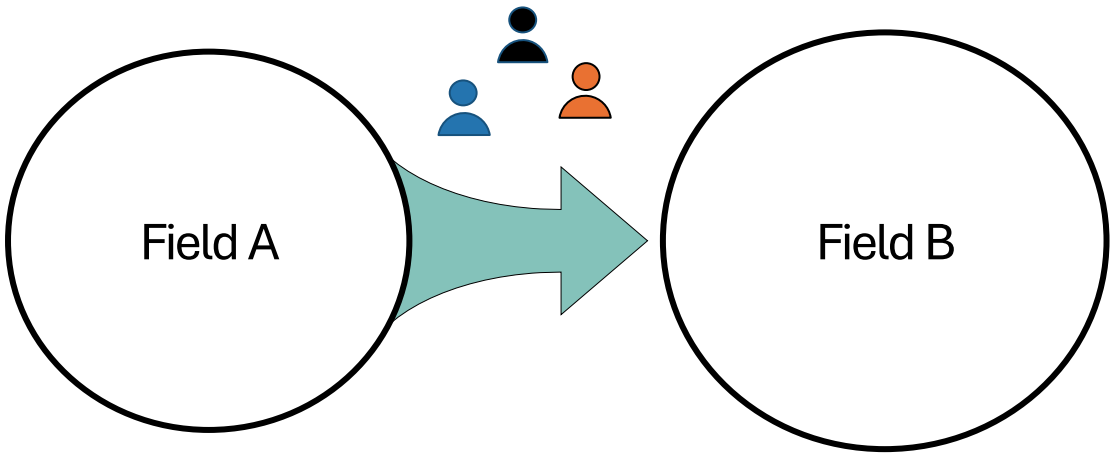


- **Distribution of research activity**
  - What scientists (collectively) do
  - **Where scientists focus their research and how participation varies across topics and fields**
  - Some fields grow by attracting more participation, while others lose it.

- **Structure of scientific knowledge**
  - What scientists (collectively) know
  - **How scientific knowledge components are connected to one another**
  - Previously separated areas of knowledge may become connected.

Research Landscape Changes as *Scientists Reorient*

> Scientists **shift their participation across topics and fields**, reshaping ***the distribution of research activities***.



- **Research reorientation**
  - **A shift in research topics or fields, relative to one's prior work**
  - Scientists choose and switch topics for various reasons, such as rewards, social context, and path dependence (Xiang et al.)

- **Scientists' reorientation reshapes the distribution of research activities**
  - Research areas emerge, grow, and decline, with participants moving toward new areas (Mulkay et al.)

Xiang et al., *Quant. Sci. Stud.* 6 (2025): 623–651; Mulkay et al., *Sociol* 9.2 (1975): 187-203

Scientific Change Appears in Two Aspects

Scientific Change

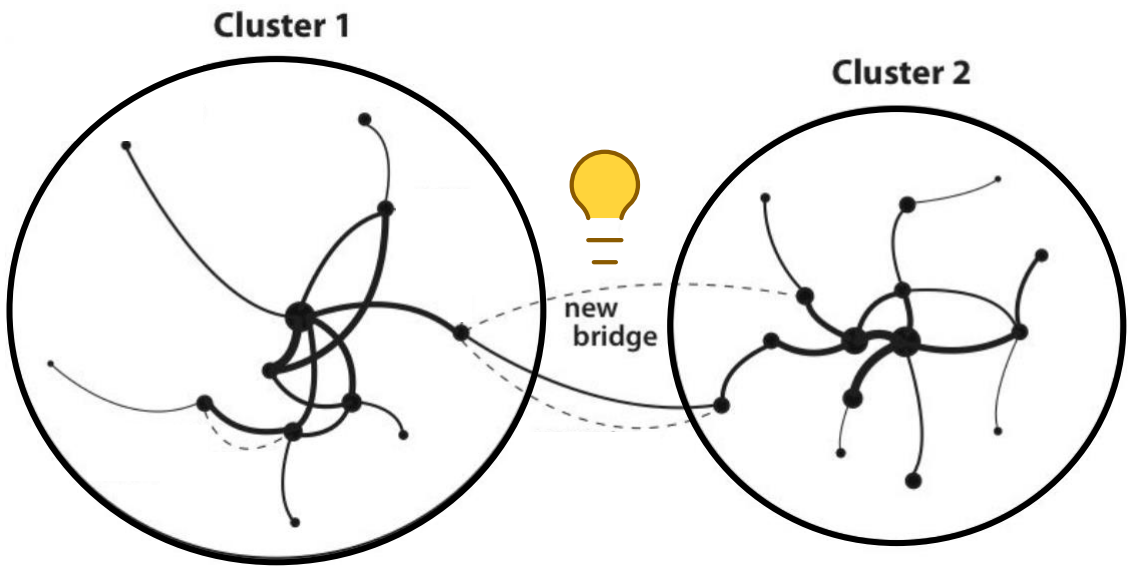
Shifts in research activities	Changes in knowledge structure

- Distribution of research activity
  - What scientists (collectively) do
  - **Where scientists focus their research and how participation varies across topics and fields**
  - Some fields grow by attracting more participation, while others lose it.

- **Structure of scientific knowledge**
  - What scientists (collectively) know
  - **How scientific knowledge components are connected to one another**
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Knowledge Structure Changes as *Scientists Recombine*

> Scientists **create new connections among existing knowledge**, reshaping ***the structure of scientific knowledge***.



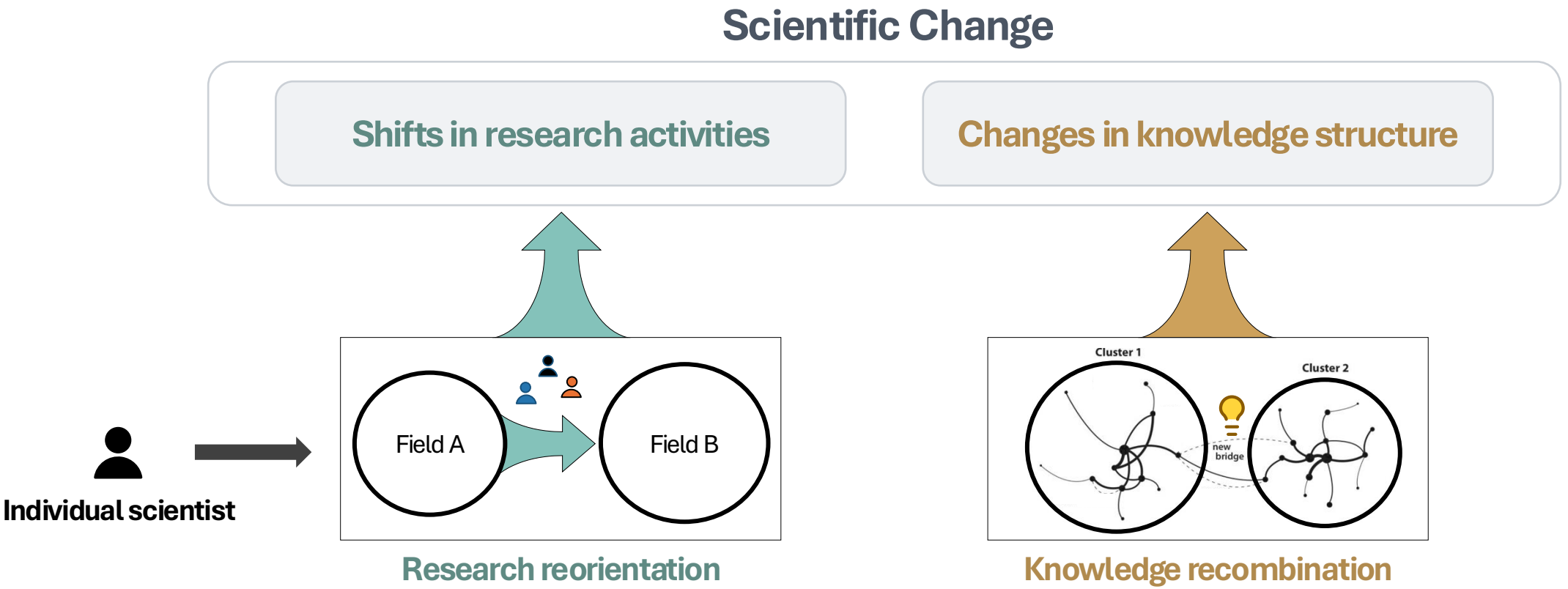
- **Knowledge recombination**
  - **Combining existing knowledge components**
  - Innovation combines components in a new way (Schumpeter)

- **Scientists' recombination reshapes the structure of scientific knowledge**
  - Research can reinforce existing connections or bridge previously separated domains (Foster et al.)

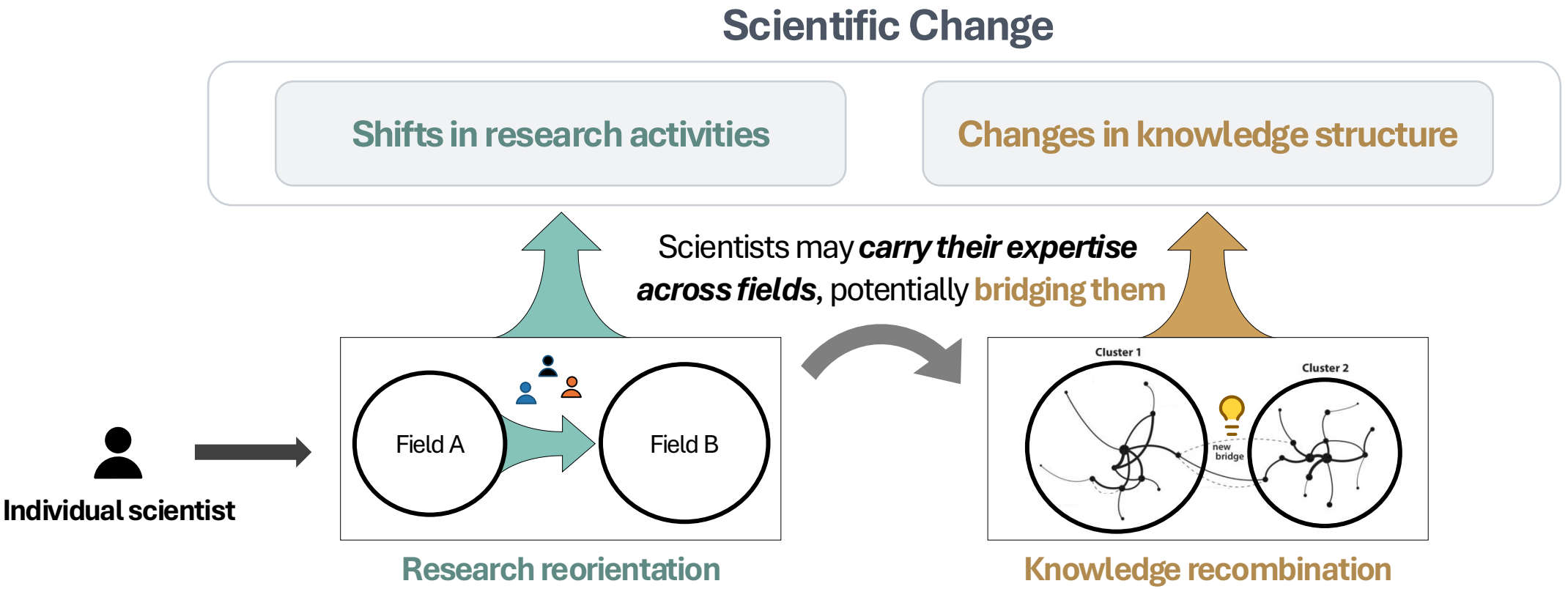
Fig: Modified from Foster et al., *Am. Sociol. Rev.* 80.5 (2015): 875-908.

Schumpeter, *The Theory of Economic Development* (1934)
Foster et al., *Am. Sociol. Rev.* 80.5 (2015): 875-908.

Scientists' Behavior in Scientific Change



Scientists' Behavior in Scientific Change



Away from Expertise, Toward Novel Combinations

> **Exploitation-exploration trade-off** (March)
> : Exploiting proximal opportunities – exploring new possibilities

Research reorientation

Less expertise

(-)

Decreased value

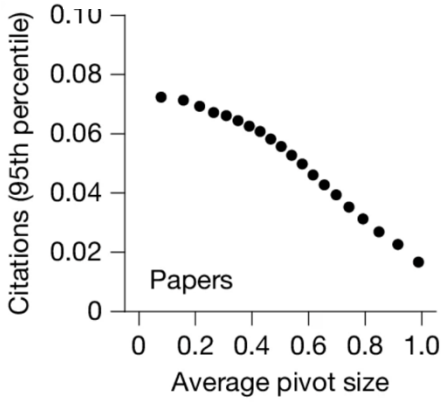


Fig: Hill et al., *Nat.* 642, 999 (2025).

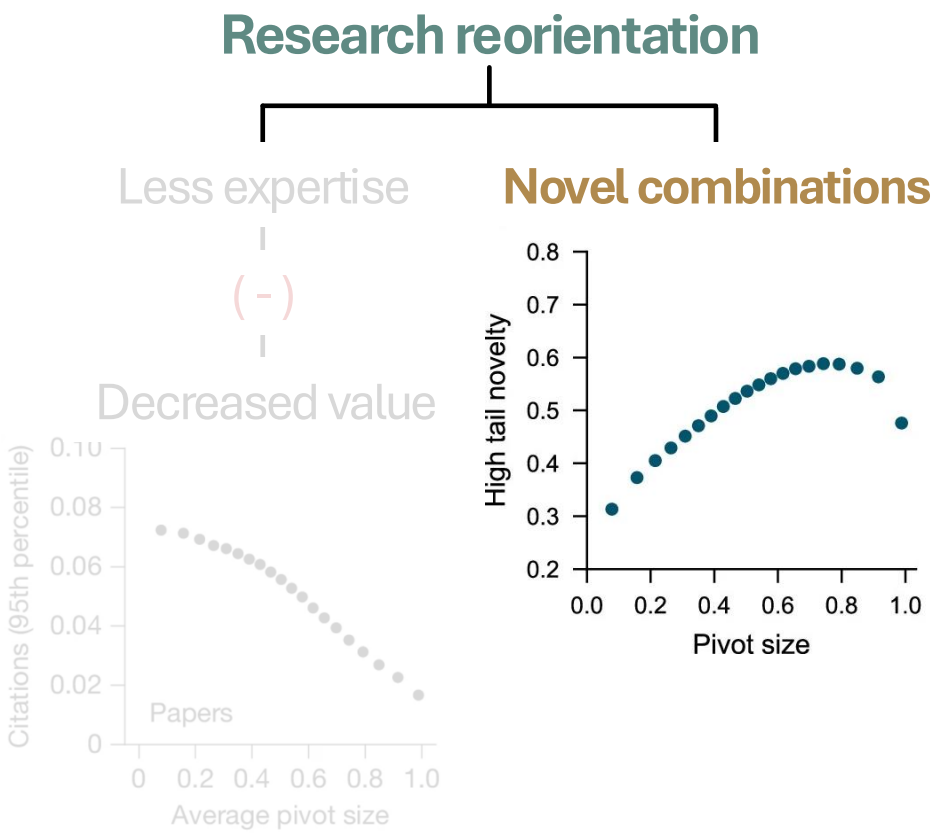
- **Exploring new fields and inventive output** (Arts & Fleming)
  - lack deep knowledge and skills in the new field and need time to reach the frontier of knowledge
- **Pivot penalty in research** (Hill et al.)
  - as they shift to new areas personally, are equipped for novelty but limited in their relevant or conventional expertise

- ***But where do the novel combinations come from?***
  - Reorientation is measured per paper.
  - Which components form the novel combinations is unseen.

March, *Organ. Sci.* 2, 71 (1991); Arts & Fleming, *Organ. Sci.* 29, 1074 (2018).

Away from Expertise, Toward Novel Combinations

> **Exploitation-exploration trade-off** (March)
> : Exploiting proximal opportunities – exploring new possibilities



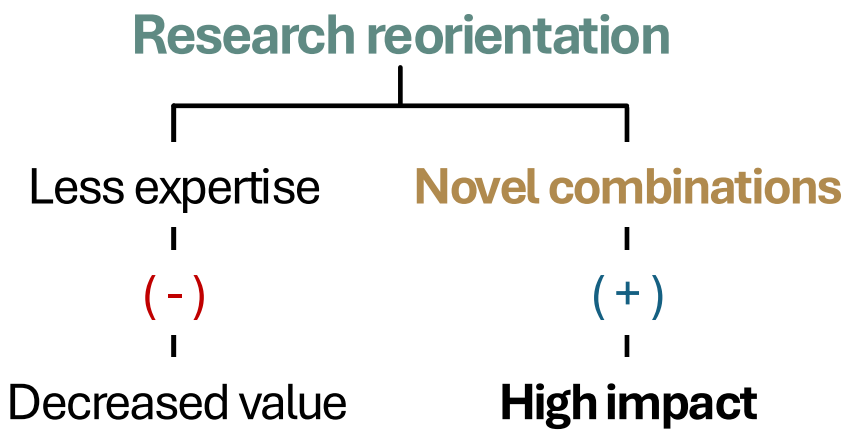
- **Exploring new fields and inventive output** (Arts & Fleming)
  - lack deep knowledge and skills in the new field and need time to reach the frontier of knowledge
  - a greater diversity in knowledge affords more possibilities
- **Pivot penalty in research** (Hill et al.)
  - as they shift to new areas personally, are equipped for novelty but limited in their relevant or conventional expertise
- ***But where do the novel combinations come from?***
  - Reorientation is measured per paper.
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Fig: Hill et al., *Nat*. 642, 999 (2025).

March, *Organ*. *Sci*. 2, 71 (1991); Arts & Fleming, *Organ*. *Sci*. 29, 1074 (2018).

Novel Combinations Without Value?

> Research reorientation brings **novel combinations** yet **lowers impact.**

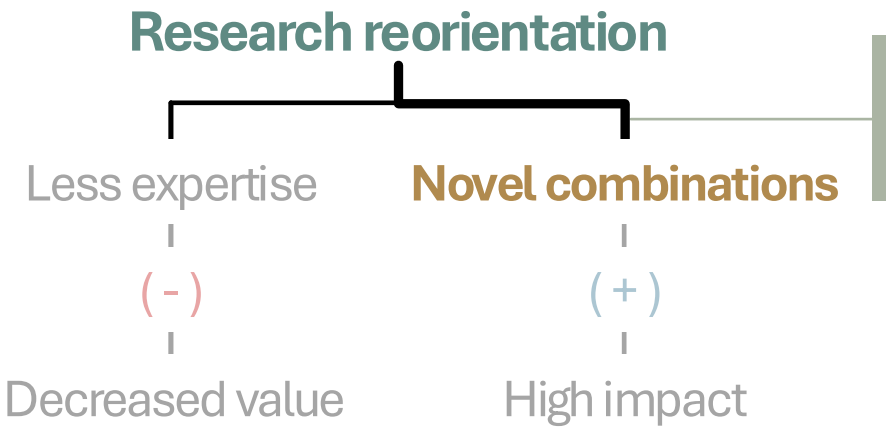


- **Mixed evidence on the value of novel combinations**
  - Lower value on average, but higher variability (Fleming)
  - Higher impact in the long run, delayed recognition (Wang et al.)
  - Pays only when grounded in established knowledge (Uzzi et al., Arts & Veugelers)

- ***But what are these novel combinations worth?***
  - The penalty mixes **what is lost from leaving expertise** with **what novelty may add**.

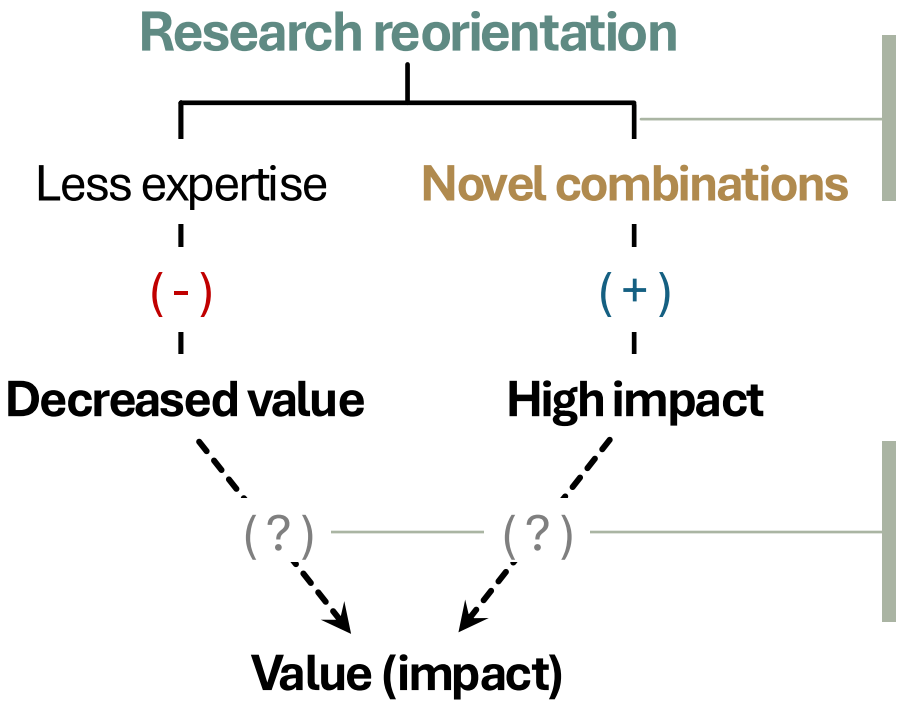
Fleming, *Manag. Sci. 47*, 117 (2001); Wang et al., *Res. Policy* 46, 1416 (2017); Uzzi et al., *Sci.* 342.6157 (2013): 468-472; Arts & Veugelers, *Ind. Corp. Change.* 24, 1215 (2015).

Research Gap: Where do novel combinations come from?



RQ1. When scientists **reorient**,
where do their **novel combinations** come from?

Research Gap: What are these novel combinations worth?

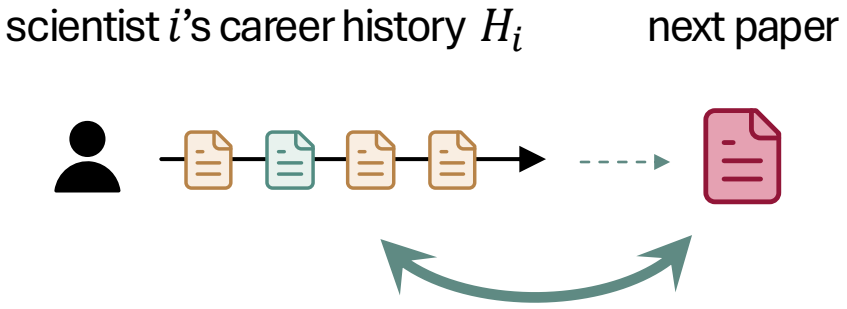


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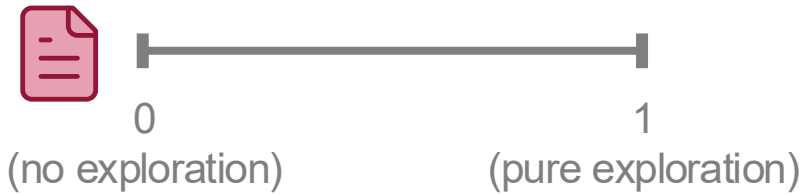
RQ2. When scientists **reorient**,
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Measurement: Decomposing Reorientation

> Measuring **research reorientation** at the component level:
> how ***familiar or distant each component is to the scientist***.

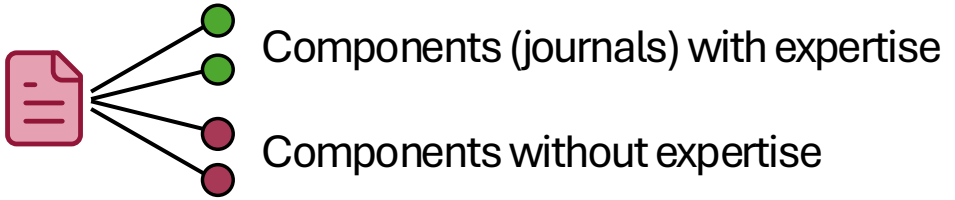


- Prior studies has measured reorientation only at the paper level.



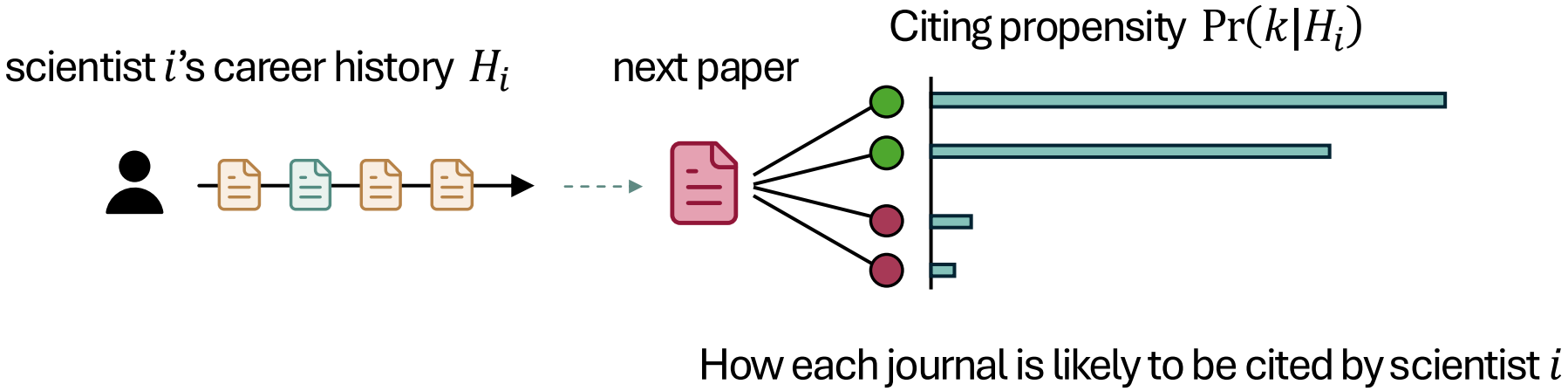
- To analyze recombination, we need **a component-level measure of reorientation.**

next paper's references $\{k_a, k_b, \ldots\}$



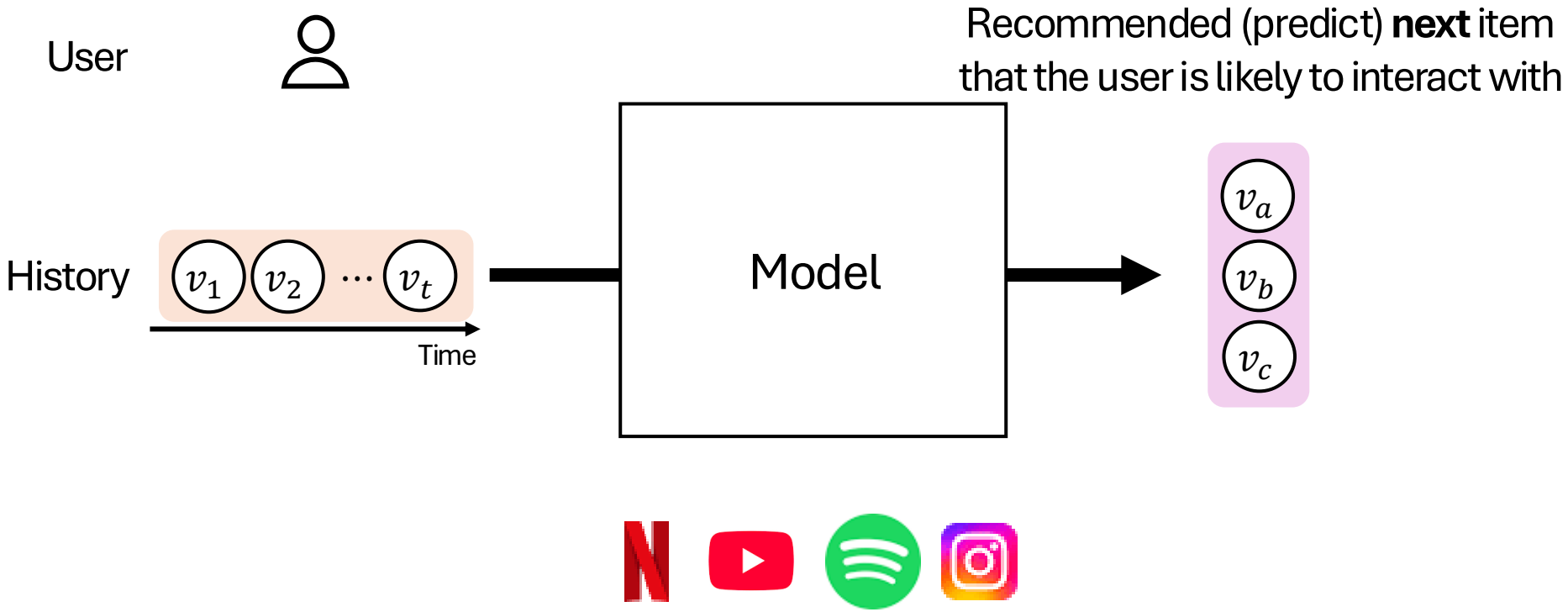
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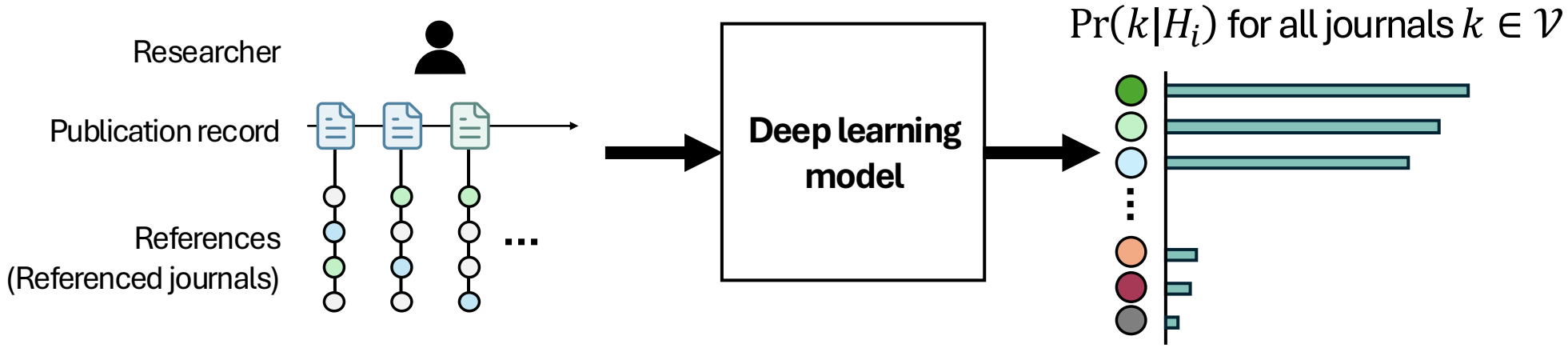
Measurement: Sequential Recommendation

> Deep learning > Recommender system > **Sequential recommendation**



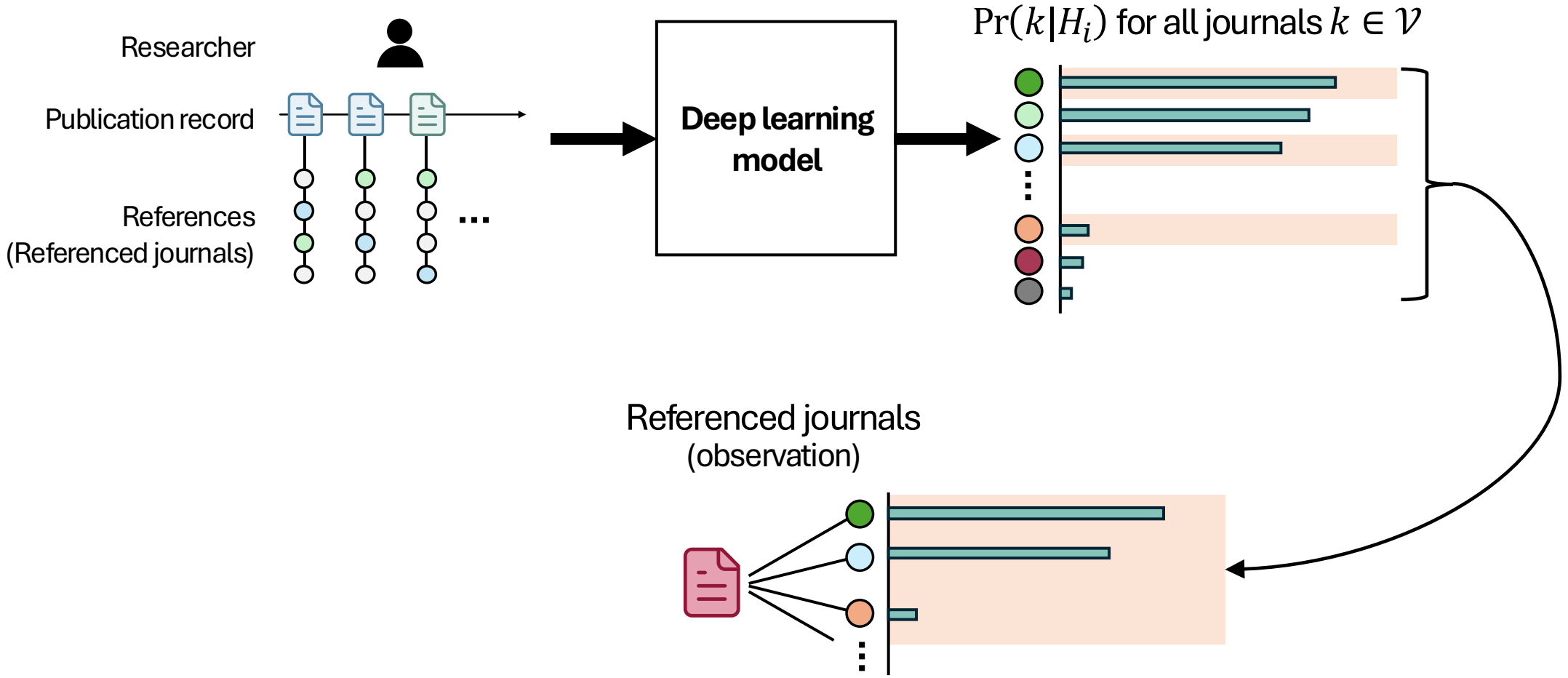
Measurement: Predicting References in the Next Paper

> Train the model to predict ***which journals the researcher is likely to cite*** in their next paper.



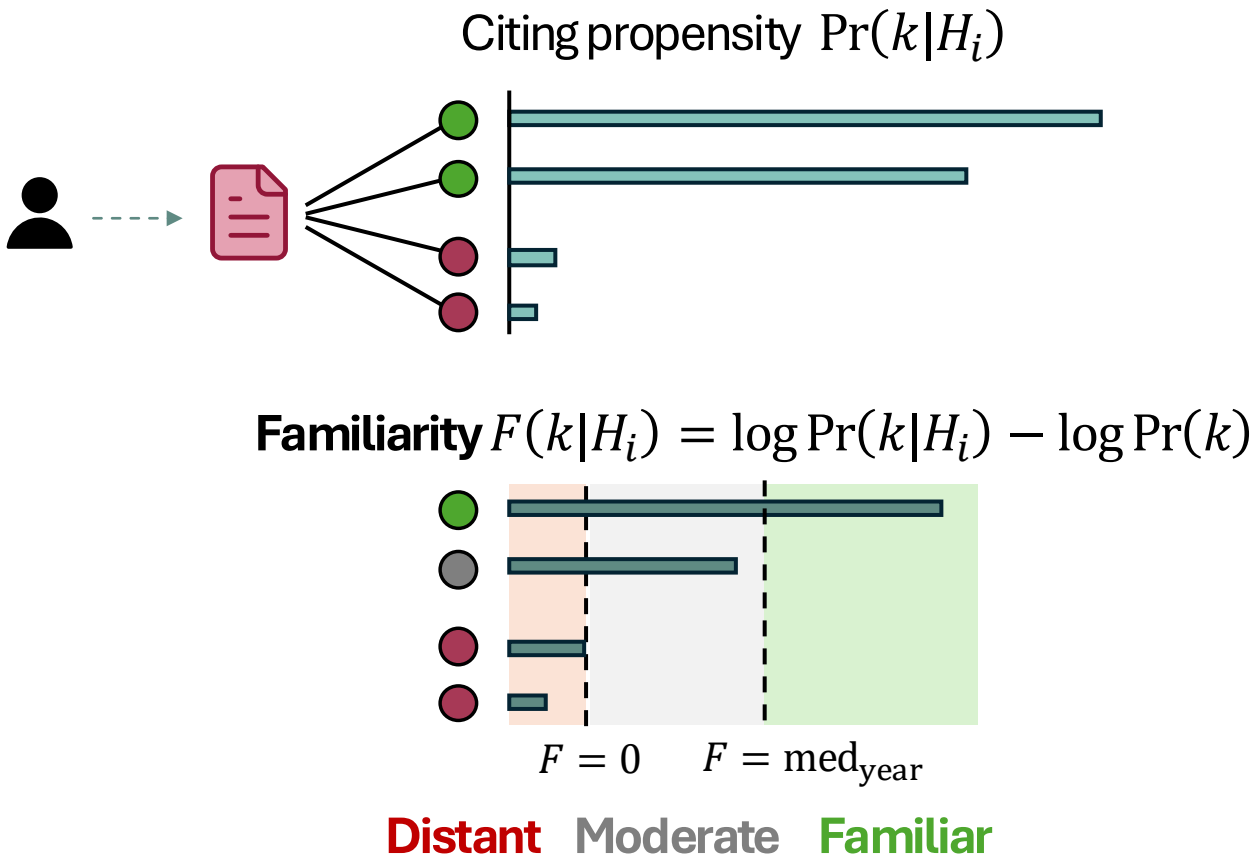
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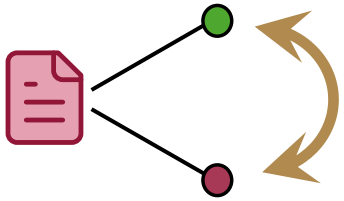


	[%]	mean	std	med
Familiar references share		61.5	28.5	67.7
Moderate references share		28.0	22.0	23.5
Distant references share		10.5	17.9	4.5

First-author papers, 1980–2019.
Familiar: F≥year median; distant: F<0.

Measurement: Novelty of Combinations

> Measuring **novelty of combinations**:
> how ***rarely*** each components are ***used together***.



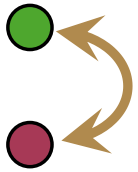
*How **rarely** they are used together?*

$$\mathrm{PMI}(k_a, k_b) = \log \frac{\Pr(k_a, k_b)}{\Pr(k_a)\Pr(k_b)}$$

$> \mathbf{0}$: Typical
$< \mathbf{0}$: **Atypical**

- $\Pr(k_a)$: Among 10,000 papers, cited 500 times ➔ 0.05
- $\Pr(k_b)$: Among 10,000 papers, cited 200 times ➔ 0.02

➔ Expected co-citation: 0.05*0.02=0.001

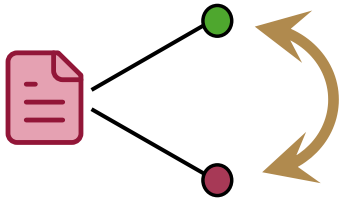


$\Pr(k_a, k_b)$: Among 10,000 papers, co-cited **2** times ➔ 0.0002

$$\mathrm{PMI}(k_a, k_b) = \log \frac{0.0002}{0.001} < \mathbf{0}$$

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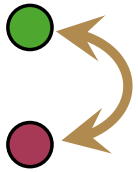
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➔ Expected co-citation: 0.05*0.02=0.001

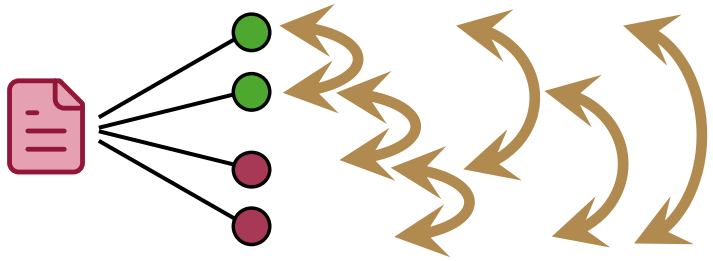


$\Pr(k_a, k_b)$: Among 10,000 papers, co-cited **100** times ➔ 0.01

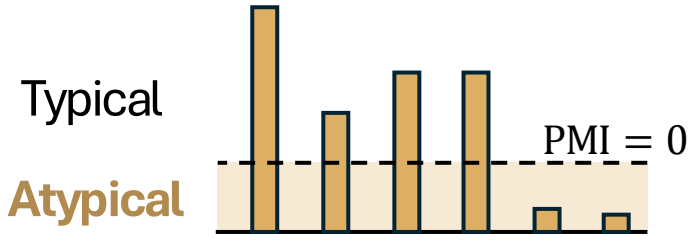
$$\mathrm{PMI}(k_a, k_b) = \log \frac{0.01}{0.001} > \mathbf{0}$$

Measurement: Novelty of Combinations

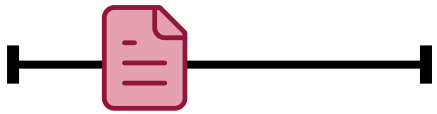
> Measuring **novelty of combinations**:
> how ***rarely*** each components are ***used together***.



	mean	std	med
Atypical combinations share [%]	10.8	11.5	7.4



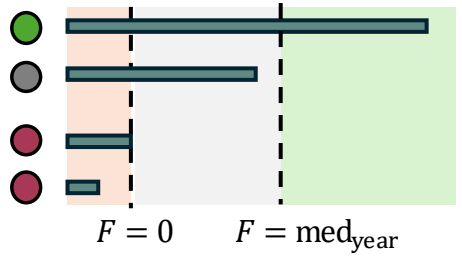
Conventionality of the paper := $\mathrm{med}(\mathrm{PMI})$
: how commonly its references are combined

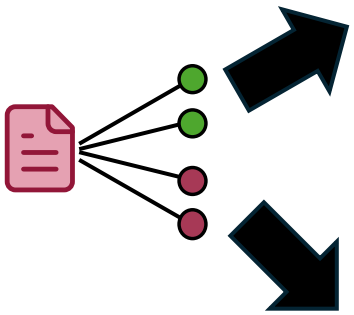


Share of **atypical** combinations

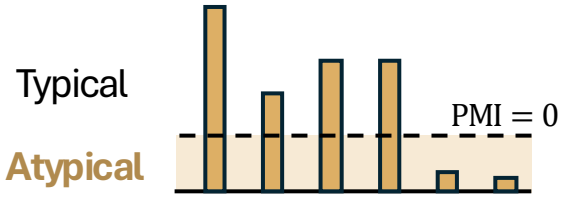
First-author papers, 1980–2019.
Atypical: reference pairs with PMI < 0. Conventionality: median of reference pairs' PMI.

Measurement: Reorientation and Combinations





- Share of familiar references
- Share of moderate references
- Share of distant references



- Share of atypical combinations
- Conventionality := $\text{med}(\text{PMI})$

Papers	Atypical share [%]	Conventionality [percentile]
Familiar share <= 67.7% (med)	15.3	40
Familiar share > 67.7%	6.3	60
Moderate share <= 23.5% (med)	8.0	58
Moderate share > 23.5%	13.7	42
Distant share <= 4.5% (med)	5.4	60
Distant share > 4.5%	16.3	40

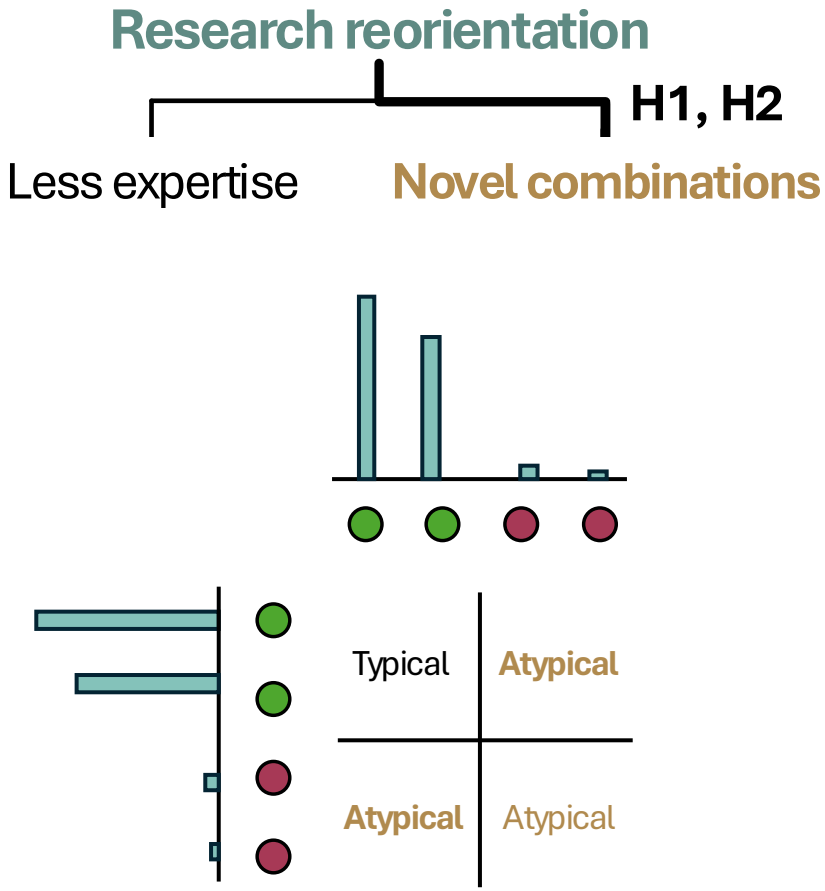
Papers	References share [%]		
	Familiar	Moderate	Distant
Atypical share = 0%	81.0	15.3	3.8
Atypical share 0~7.4% (med)	66.6	26.7	6.7
Atypical share > 7.4%	51.4	33.2	15.4

Where Do Novel Combinations Come From?

> **RQ1.** When scientists **reorient**, where do their **novel combinations** come from?

- **H1.** Papers that draw on *more* distant components contain *more* atypical combinations.
  - (Arts & Fleming; Hill et al.)

- **H2.** The combinations become *more* atypical when distant components are paired with familiar components.
  - The interaction of search depth and search scope may lead to solutions of greater novelty (Katila & Ahuja; Schilling & Green)



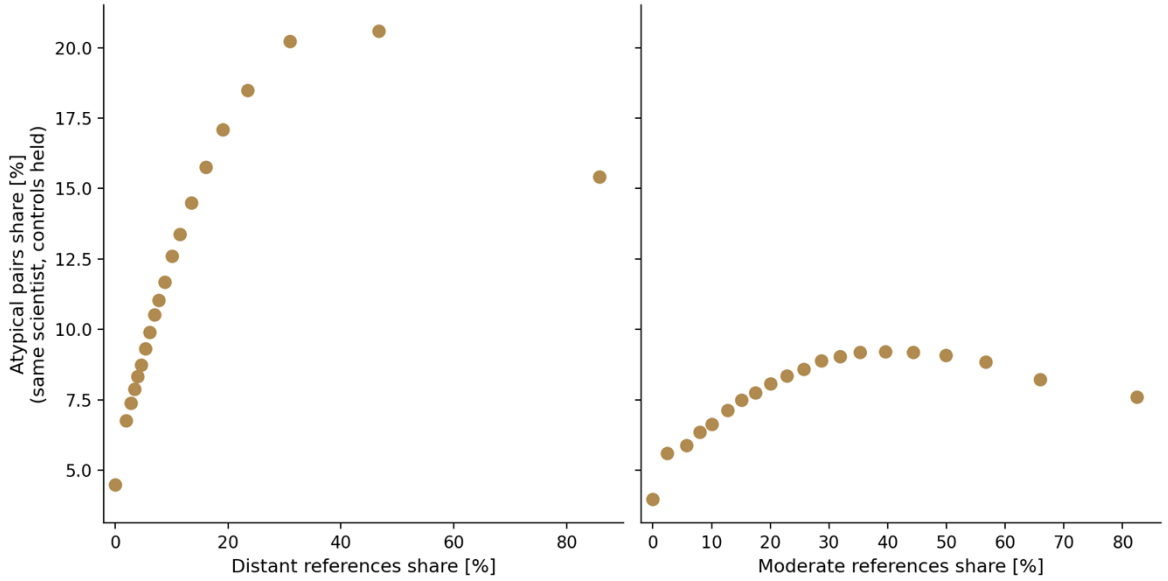
Arts & Fleming, *Organ. Sci.* 29, 1074 (2018); Hill et al., *Nat.* 642, 999 (2025).
Katila & Ahuja, Acad. Manage. J. 45, 1183 (2002); Schilling & Green, *Res. Policy 40*, 1321 (2011)

Novelty Emerges from Combining Distant and Familiar Knowledge

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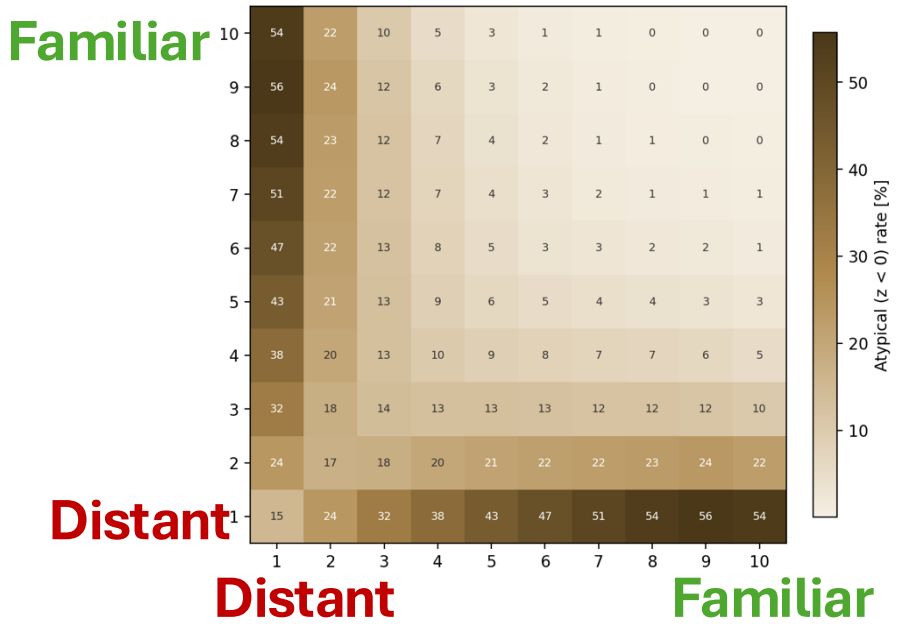
- **H1.** Papers that draw on *more* distant components contain *more* atypical combinations. ➔ **Supported**

$$\text{Share}(\text{atypical pairs}) \sim \text{Share}(\text{distant}) + \text{Share}(\text{moderate})$$



- **H2.** The combinations become *more* atypical when distant components are paired with familiar components. ➔ **Supported**

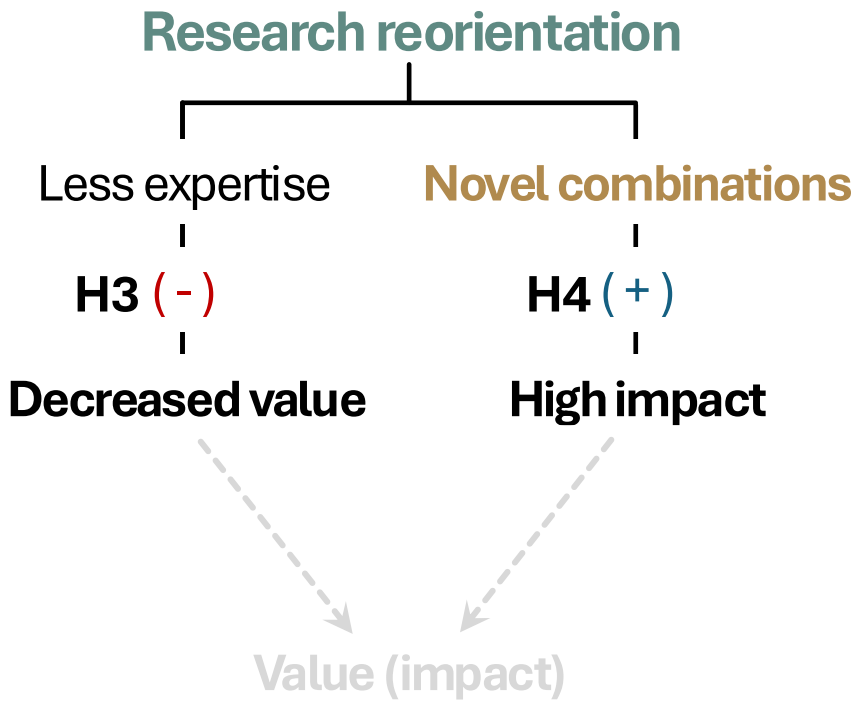
$$\text{PMI}(k_a, k_b) \sim F(k_a|H_i) + F(k_b|H_i)$$



Reorientation and Novelty: Competing Links to Impact

> **RQ2.** When scientists **reorient**, do **novel combinations** mitigate the penalty of losing expertise?

- **H3.** Papers with *more* distant components are *less* likely to be high-impact.
  - (Arts & Fleming; Hill et al.)
- **H4.** Papers with *more* atypical combinations are *more* likely to be high-impact.
  - (Uzzi et al.; Schilling & Green)



Arts & Fleming, *Organ. Sci.* 29, 1074 (2018); Hill et al., *Nat.* 642, 999 (2025).
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Distance Predicts Lower Impact; Novelty Predicts Higher Impact

RQ2. When scientists **reorient**, do **novel combinations** mitigate the penalty of losing expertise?

- **H3.** Papers with *more* distant components are *less* likely to be high-impact.
  ➔ **Supported**

- **H4.** Papers with *more* atypical combinations are *more* likely to be high-impact.
  ➔ **Supported**

$$\text{Top5\% citation} \sim \text{Share(distant)} + \text{Share(moderate)} + \text{Share(atypical pairs)} + \text{Conventionality}$$

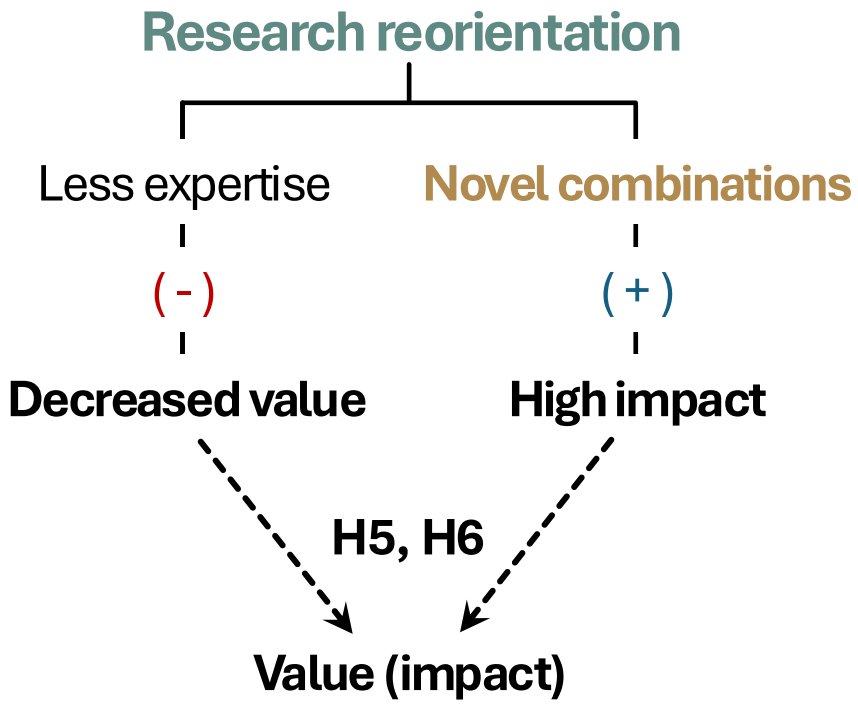
Y: Top 5% citation in field-year	**M3**	**M4**
Distant references share (+1%p)	**-0.0084*** **	**-0.0119*** **
Moderate references share (+1%p)	+0.0093***	+0.0108***
Atypical pairs share (+1%p)		**+0.0653*** **
Conventionality (+1 percentile)		+0.0219***
FE	(Field×year)-FE, Author-FE	
R^2	0.33496	0.33516
Within R^2	0.03187	0.03215
Observations	6,962,773	

Linear probability model.
Controls: reference count, number of journals in references, reference age, team size.
***: p<0.001, *:p<0.05. Author-clustered standard errors. First-author papers 1980–2019.

Can Novelty Offset the Reorientation Penalty?

> **RQ2.** When scientists **reorient**, do **novel combinations** mitigate the penalty of losing expertise?

- **H5.** Atypical combinations *offsets* the negative relation in **H3**.
  - (**H3.** *more* distant, *less* high-impact papers)
  - Exploration creates opportunities for valuable recombination alongside the difficulty of moving beyond one's expertise.



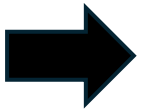
Novelty Partly Offsets the Reorientation Penalty

> **RQ2.** When scientists **reorient**, do **novel combinations** mitigate the penalty of losing expertise?

- **H5.** Atypical combinations *offsets* the negative relation in **H3.** ➔ **Supported**

$$\text{Top5\% citation} \sim \text{Share(distant)} + \text{Share(moderate)} + \text{Share(atypical pairs)} + \text{Conventionality}$$

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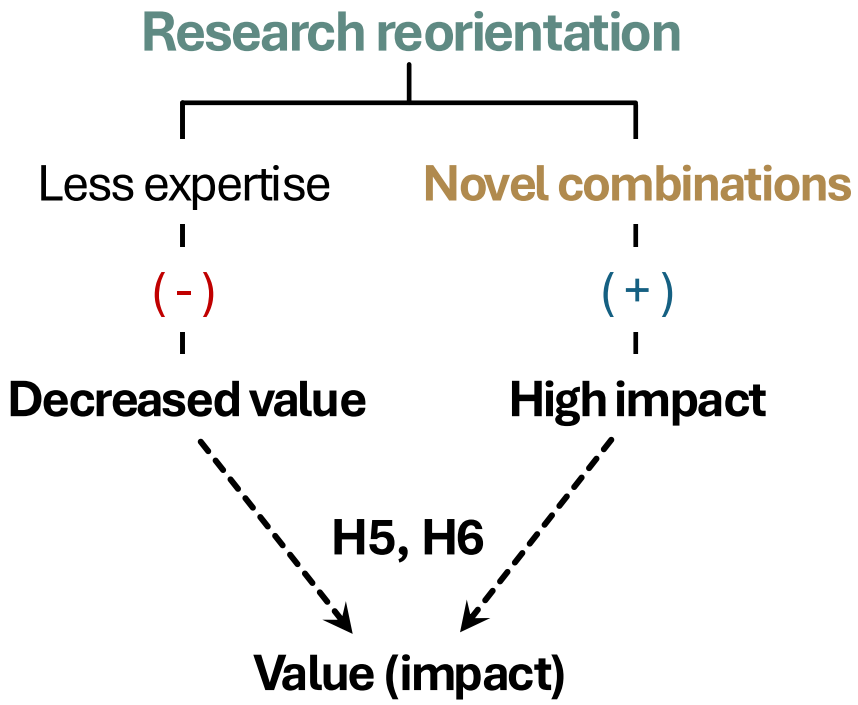
Gelbach decomposition

Coefficient of Share(distant)	**Estimate (SE)**
Coefficient in **M3**	−0.0084 (0.0009)
via higher atypicality	**+0.0129** (0.0004)
via lower conventionality	**−0.0094** (0.0003)
Coefficient in **M4**	−0.0119 (0.0009)

Does Familiar Knowledge Strengthen the Return to Novelty?

> **RQ2.** When scientists **reorient**, do **novel combinations** mitigate the penalty of losing expertise?

- **H6.** The positive relation in **H4** is *stronger* in papers with *more* familiar components.
  - (**H4.** *more* atypical, *more* high-impact papers)
  - Prior knowledge facilitates the absorption and productive use of new knowledge (Cohen & Levinthal; Katila & Ahuja)



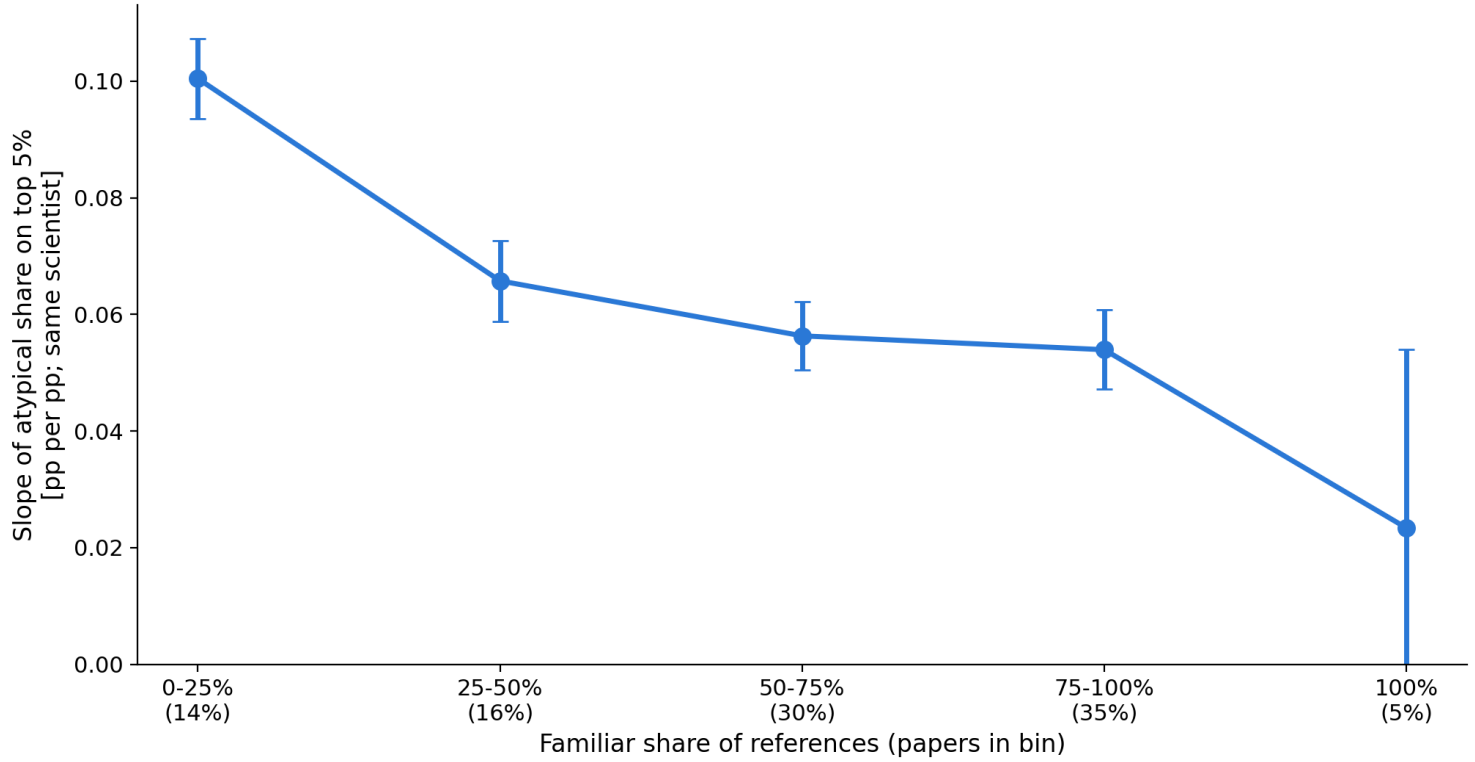
Cohen & Levinthal, Adm. Sci. Q. 35, 128 (1990); Katila & Ahuja, Acad. Manage. J. 45, 1183 (2002).

No Support for Familiarity Amplifying the Novelty Premium

RQ2. When scientists **reorient**, do **novel combinations** mitigate the penalty of losing expertise?

- **H6.** The positive relation in **H4** is *stronger* in papers with *more* familiar components.
  ➔ **Not supported**

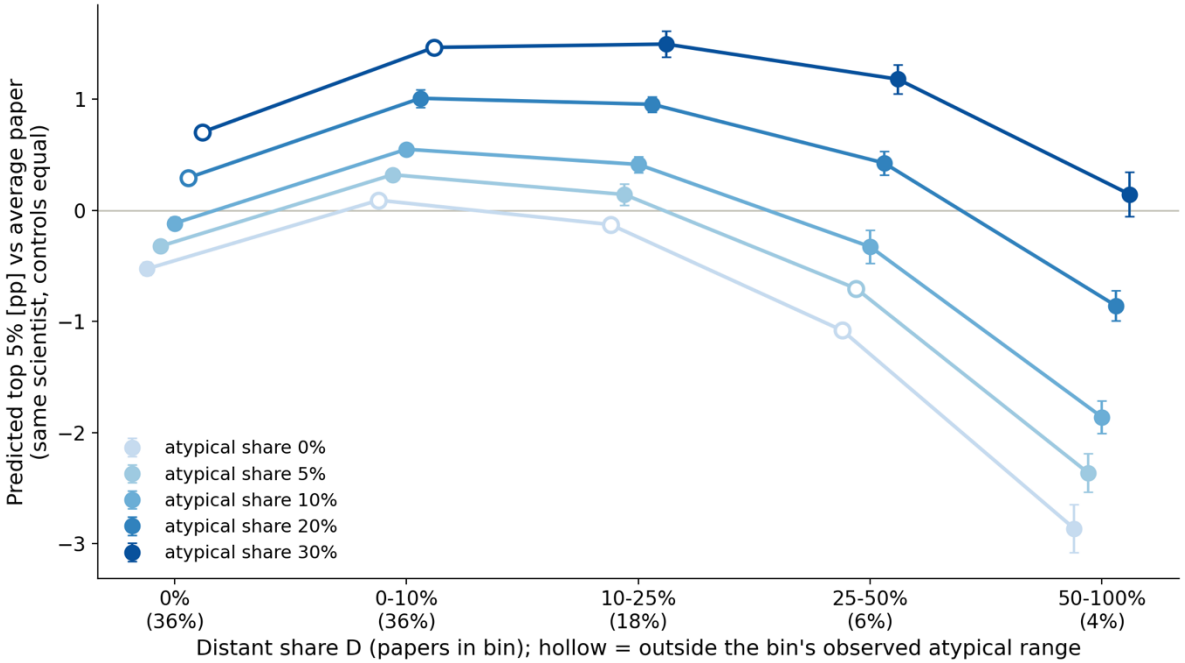
$$\text{Top5\% citation} \sim \text{Share(components)} + \text{Share(atypical pairs)} + \text{Share(atypical pairs)} \times \text{Share(components)}$$



Novelty Matters More When Scientists Move Farther

RQ2. When scientists **reorient**, do **novel combinations** mitigate the penalty of losing expertise?

- **H6.** The positive relation in **H4** is *stronger* in papers with *more* familiar components.
- ~ Expertise *strengthens* the return to novelty.
  ➔ **Not supported**



- Contrary to **H6**, the positive relation in **H4** is *stronger* in papers with *more* distant components.
- ~ Papers drawing on distant knowledge are more dependent on atypical combinations for high impact.
- ~ ***Compensatory*** **role of novelty**

Where Novelty Comes From—and When It Matters

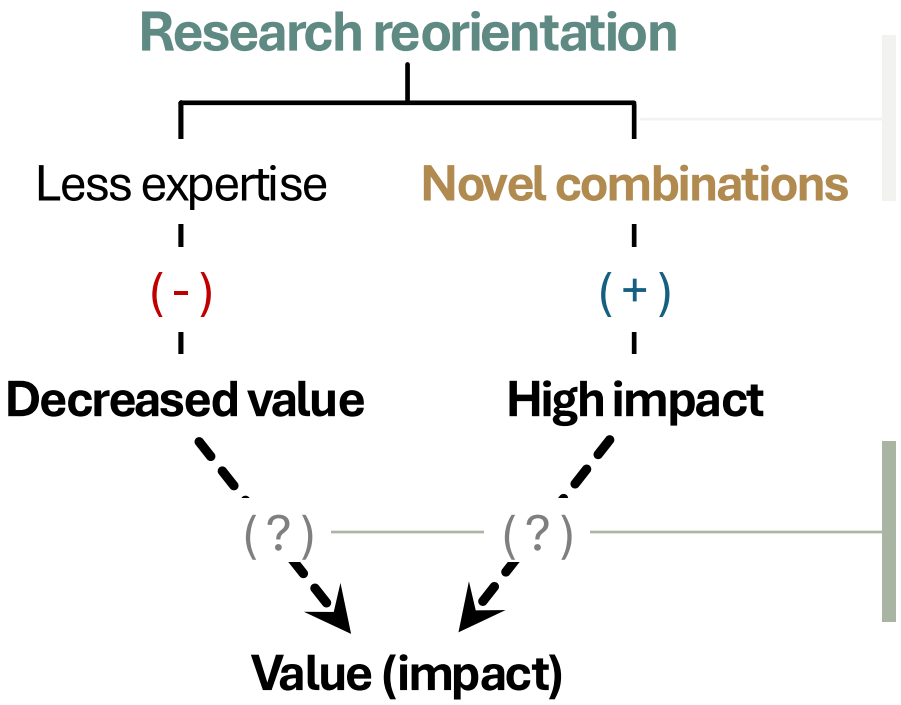


RQ1. When scientists **reorient**,
where do their **novel combinations** come from?

➔ **Novelty comes from exploring distant components while anchored in familiar ones.**

RQ2. When scientists **reorient**,
do **novel combinations** mitigate the penalty of losing expertise?

Where Novelty Comes From—and When It Matters



RQ1. When scientists **reorient**,
where do their **novel combinations** come from?

➔ **Novelty comes from exploring distant components while anchored in familiar ones.**

RQ2. When scientists **reorient**,
do **novel combinations** mitigate the penalty of losing expertise?

➔ **Novel combinations can offset the penalty.**
The farther scientists move, the more novelty matters.